

IA: MACHINE LEARNING FOUNDATIONS

SESSION 1 - Course introduction and overview of Machine learning

Adrián Sánchez Morales

AGENDA

Professor Bio

Syllabus

Program

Bibliography

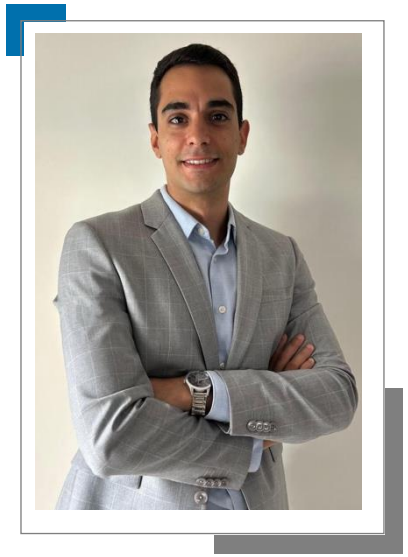
Evaluation

Machine Learning

What do you expect?



PROFESSOR BIO



Adrián Sánchez Morales

- Head of AI&BI
- Advanced Analytics Senior Manager

Capgemini

- Lead Data Scientist in Europe
- +8 years of experience

BOEING

- PhD in Artificial Intelligence
- Several papers in journals with high impact factor

Universidad Carlos III de Madrid

- Telecommunications Engineer

Universidad Politécnica de Cartagena

www.linkedin.com/in/adrián-sánchez-morales/

adrians@faculty.ie.edu

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What do you expect?



SYLLABUS

SUBJECT DESCRIPTION

- Machine Learning refers to a set of tools for making sense of complex datasets.
- In recent years, it has become a critical toolkit for everyone.
- This course will introduce the theory and practical machine learning algorithms.
- We will review the basics of the models commonly used in real problems.

SYLLABUS

SUBJECT DESCRIPTION

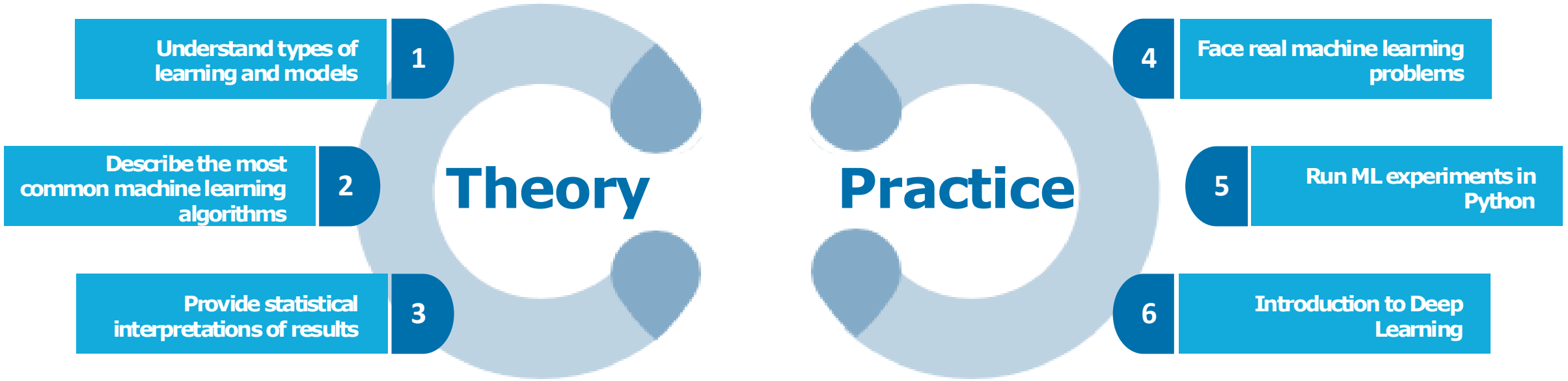
- Machine Learning refers to a set of tools for making sense of complex datasets.
- In recent years, it has become a critical toolkit for everyone.
- This course will introduce the theory and practical machine learning algorithms.
- We will review the basics of the models commonly used in real problems.

Technical Perspective

- We will consider a variety of applications, including classification, prediction, regression, clustering, modeling, and data exploration/visualization.

SYLLABUS

OBJECTIVES AND SKILLS



SYLLABUS

PREREQUISITES

Recommended:

1. Familiarity with basic concepts in linear algebra, probability and calculus
   - Introduction session
   - Individual study

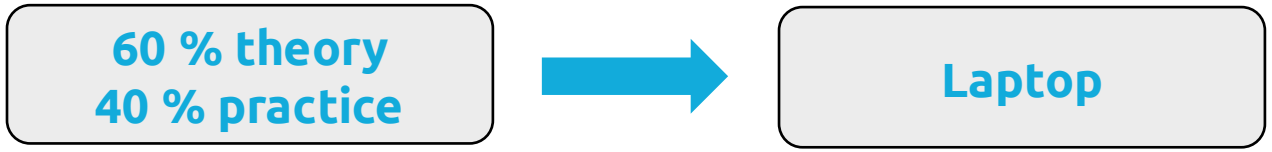


2. Prior python programming experience
   - Introduction to python environment
3. Familiarity with basic concepts in Machine Learning
   - EDA
   - Optimization models

SYLLABUS

METHODOLOGY

- We will mix theoretical lecturing and practical assignments during the sessions.



- Estimated effort:

Teaching methodology	Weighting	Estimated time a student should dedicate
Lectures	30.0 %	45 hours
Discussions	10.0 %	15 hours
Exercises	30.0 %	45 hours
Group work	20.0 %	30 hours
Other individual studying	10.0 %	15 hours
TOTAL	100.0 %	150 hours

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What do you expect?



PROGRAM

PLANNING

The following program is tentative. Although we will attempt to cover all the listed topics, the pace of the class depends on group performance.

1. **Course Introduction and overview of Machine learning**
2. **Introduction to Machine Learning**
3. **Practical session - Perceptron**
4. **Supervised Learning - Regression**
5. **Supervised Learning – Classification**
6. **Overfitting and Regularization**
7. **Practical session – Supervised Learning**
8. **Multilayer Perceptron**
9. **Tree-based Methods**
10. **Practical session – Trees**

PROGRAM

PLANNING

The following program is tentative. Although we will attempt to cover all the listed topics, the pace of the class depends on group performance.

11. **Group Work Introduction**
12. **Support Vector Machines**
13. **Practical Session – SVMs**
14. **Unsupervised Learning and Clustering**
15. **Practical Session – Clustering**
16. **Convolutional Neural Networks I**
17. **CNNs II**
18. **Practical Session – Computer Vision**
19. **Sequence Modeling I**
20. **Sequence Modeling II**

PROGRAM

PLANNING

The following program is tentative. Although we will attempt to cover all the listed topics, the pace of the class depends on group performance.

21. **Practical Session – Natural Language Processing**
22. **Advanced Real Applications**
23. **Transformers I**
24. **Transformers II**
25. **Generative AI**
26. **Retrieval Augmented Generation (RAG)**
27. **External Talk**
28. **Group Work Presentation I**
29. **Group Work Presentation II**
30. **Final exam**

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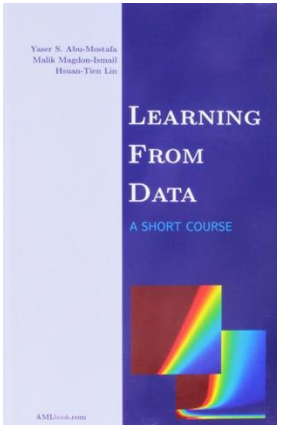
What do you expect?

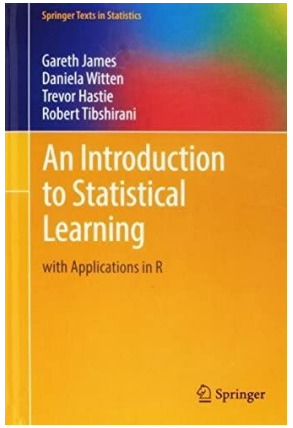


BIBLIOGRAPHY

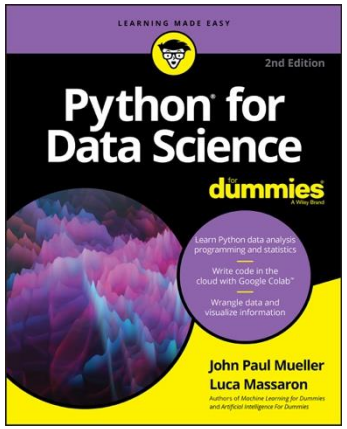
RECOMMENDED

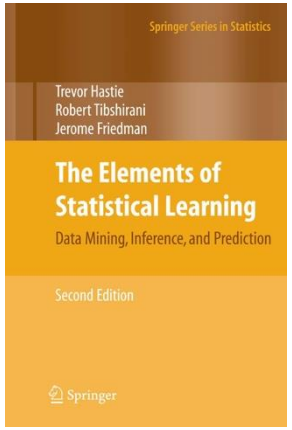
There is no formally required textbook for this course, but I will draw material heavily from these sources:

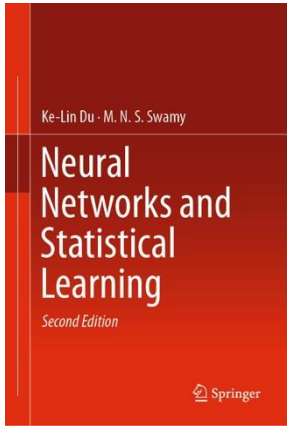




ADDITIONAL READING







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What do you expect?



EVALUATION CRITERIA

01

Class Participation

Contribution to the discussions and assignments

15 %

02

Individual Work

Resolution and presentation of a few problems and exercises



30%

DEADLINE

Defined in class

03

Group Project

Designing and implementing an algorithmic solution to a business problem

20%

DEADLINE

Sessions 28 y 29

04

Final Exam

Individual exam

35%

EVALUATION CRITERIA

IMPORTANT MESSAGES

- Assignments are INDIVIDUAL. Copying from other students is totally forbidden.
- The format for the assignment submission is a python notebook converted into PDF.
  - The document should have, in the first line, the link to an accessible notebook with the original code so that the professor can check the results.
  - Every step done should be explained and show clear results.
- The deadline for the assignments will be set in class. Late submissions are not allowed.
- Assignment submissions will be done on Blackboard. Emails are not allowed and won't be reviewed by the professor (under any circumstances).
- Generative AI is permitted for completing assignments. However, the professor may request a brief presentation of the assignment from the student at any time between the submission date and the end of the course. The student must demonstrate that they have completed the assignment and fully understand its content. If not, it'll result in a failing grade.

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MACHINE LEARNING

SUMMARY

- How can we
  1. Learn effective info from data?
  2. Apply mathematical models to practical inference problems?
- Applications: classification, prediction, regression, clustering, modeling…
- Our approach: Machine Inference
- How to reason about and work with probabilistic models to help us make inferences from data.

MACHINE LEARNING

WHAT IS MACHINE LEARNING?

- What does “learn” mean?

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- What does “learn” mean?

“Gain or acquire knowledge of or skill in (something) by study, experience, or being taught”



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- What does “learn” mean?

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- How do we learn that this is a tree?





MACHINE LEARNING

WHAT IS MACHINE LEARNING?

- What does “learn” mean?

“Gain or acquire knowledge of or skill in (something) by study, experience, or being taught”

- How do we learn that this is a tree?



A perennial plant with an elongated stem, or trunk, supporting leaves or branches



MACHINE LEARNING

WHAT IS MACHINE LEARNING?

- What does “learn” mean?

“Gain or acquire knowledge of or skill in (something) by study, experience, or being taught”

- How do we learn that this is a tree?

Examples!



- A good definition of learning for this course:

“using a set of examples to *infer* something about an underlying process”

MACHINE LEARNING

WHY LEARN FROM DATA?

- Traditional signal processing is “top down”

Given a model for our data → **derive the optimal algorithm**

- A learning approach is ”bottom up”

Given some examples → **derive a good algorithm**

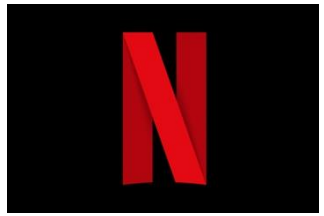
- Challenge??

Sometimes, a good model is really hard to derive from first principles

MACHINE LEARNING

EXAMPLE OF LEARNING

- The Netflix prize: Predict how a user will rate a movie



10% improvement = $1 million prize



MACHINE LEARNING

EXAMPLE OF LEARNING

- The Netflix prize: Predict how a user will rate a movie



10% improvement = $1 million prize

- Find some patterns:
  - Users do not assign ratings completely at random



  - Imagine that you have lots and lots of data:
    - We know how a user has rated other movies
    - We know how other users have rated this and other movies

MACHINE LEARNING

EXAMPLES OF LEARNING

- Recommendation Systems
- Handwritten text recognition/OCR
- Speech recognition
- Machine translation
- Face recognition
- Activity recognition
- Time series forecasting (traffic, weather, markets, etc.) • Search
- Spam filtering
- Fraud detection
- Medical diagnosis

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What do you expect?




What
do you
EXPECT


THANK YOU